MASTER'S DEGREE IN PHARMACEUTICAL BIOTECHNOLOGY

From Genome to Drug – Exploring Bioinformatics & Omics Sciences

Lesson V - Proteomics Analysis

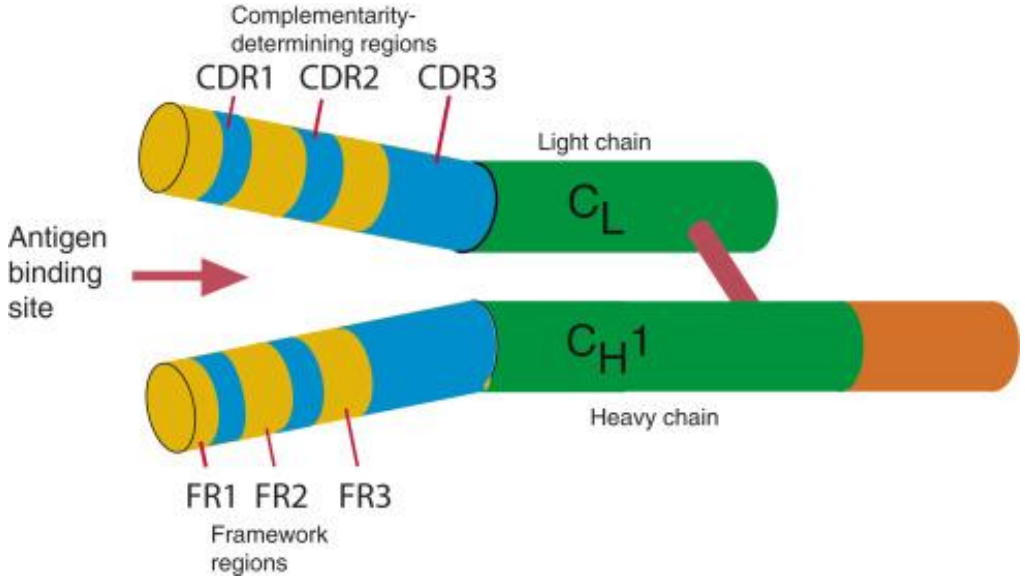
Proteomics – Intro and Protein Visualisation

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Learning Objectives — Lesson V

- Structural organization of a monoclonal antibody and locate its variable regions
- Map the amino acid residues that form the ipilimumab-CTLA-4 binding
- Identify CDRs in the Ipilimumab Fab fragment
- Visualise the interaction between an antibody and its Ag

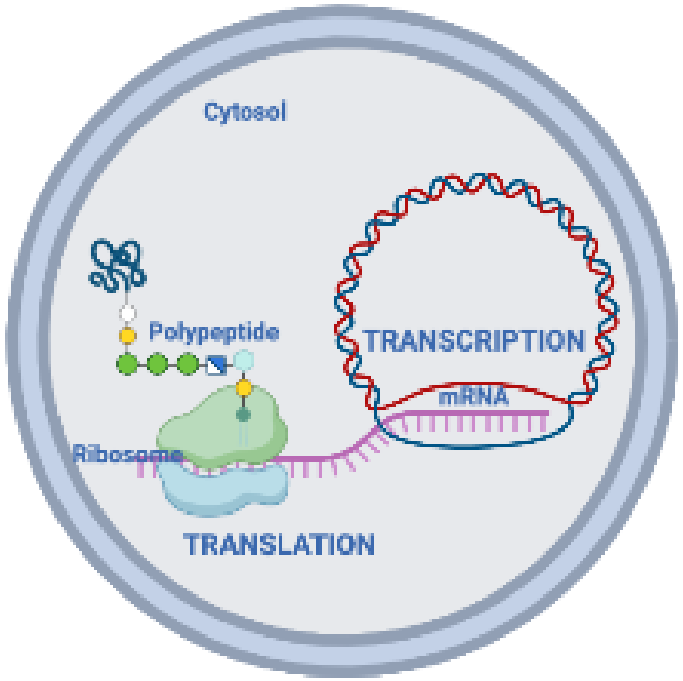


Translation

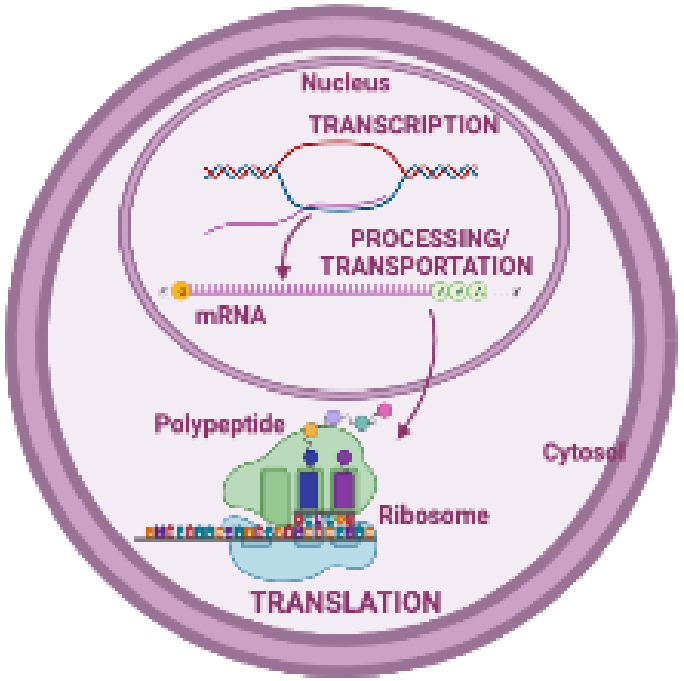
- Almost the entire genome codes for proteins.
- Genes with related functions are organized into OPERONS and are therefore transcribed in a coordinated manner from a single promoter.

POLYCISTRONIC

PROKARYOTIC CELL



EUKARYOTIC CELL



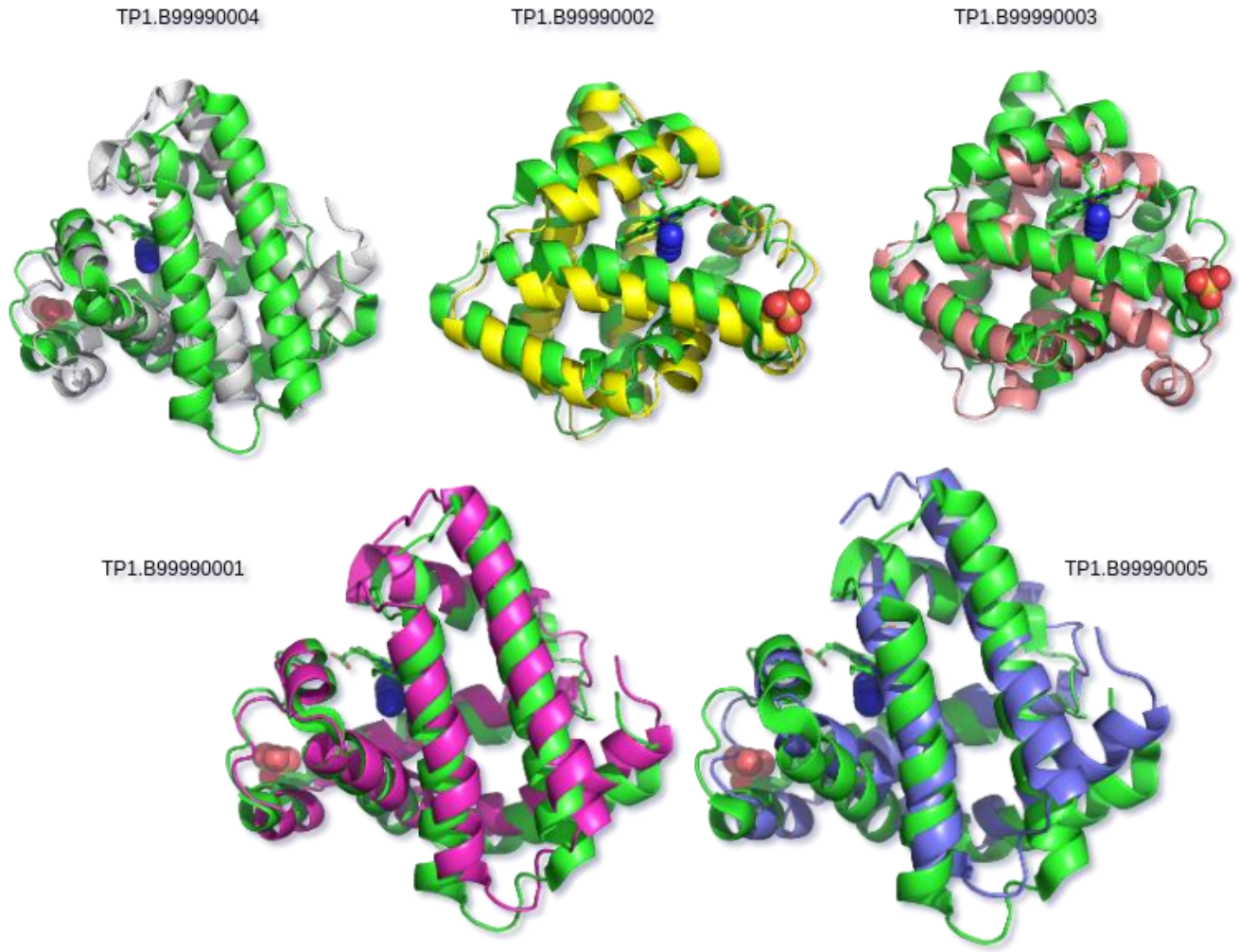
- The percentage of the genome translated into proteins is generally quite low (about 1–2% in *Homo sapiens*)

PREDOMINANTLY MONOCISTRONIC

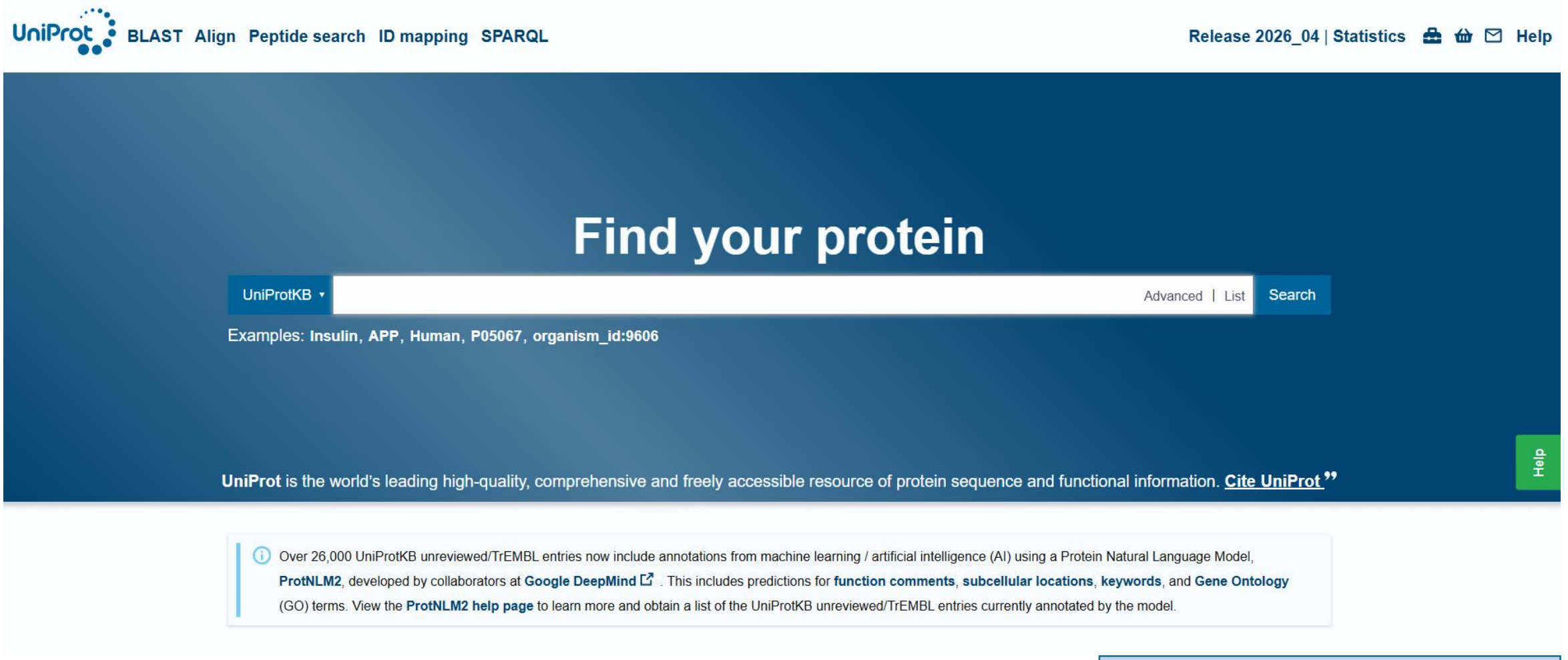
ISOFORMS (various mRNAs)

Functional Annotation of Proteins

- If two proteins are similar in sequence, and thus presumably **homologous** (sharing a common evolutionary origin), they generally have **similar** functions and **3D structures**.

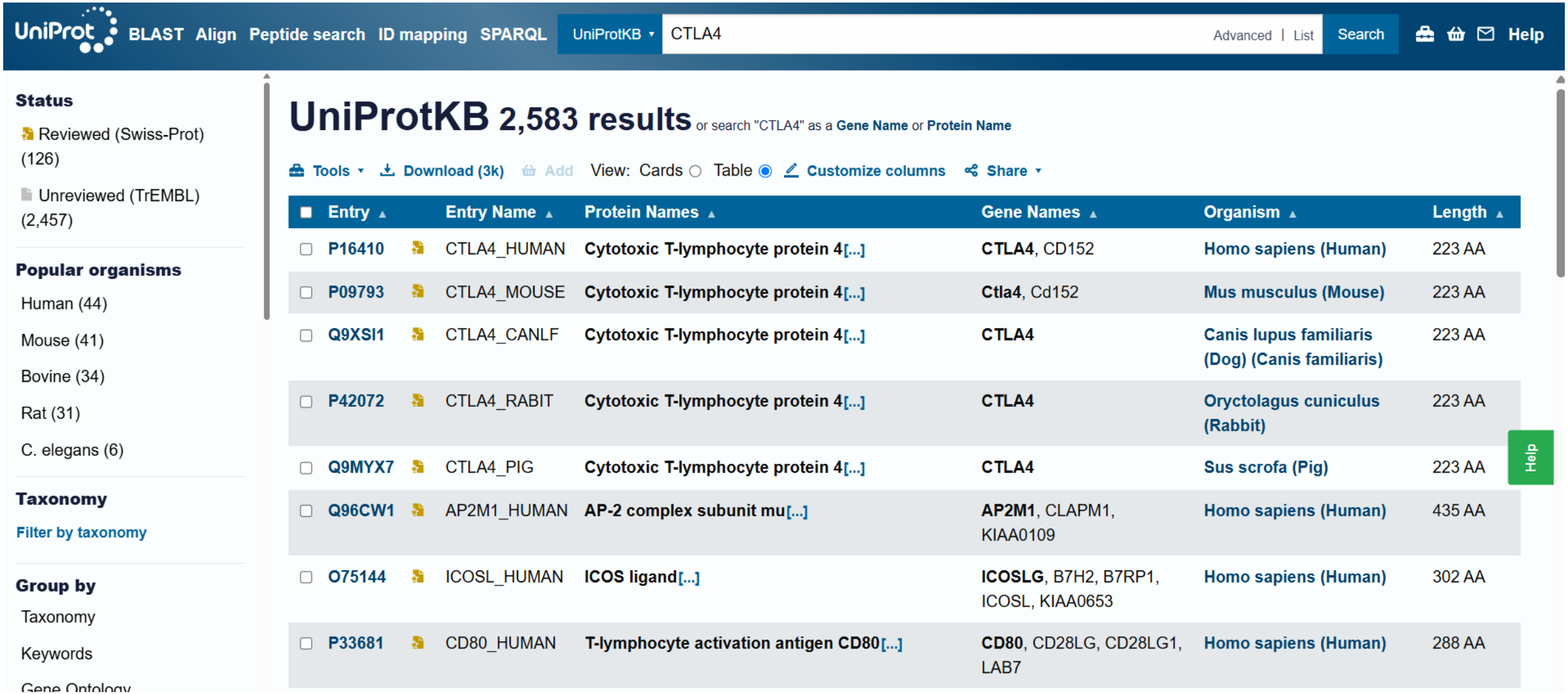


Functional Annotation of Proteins

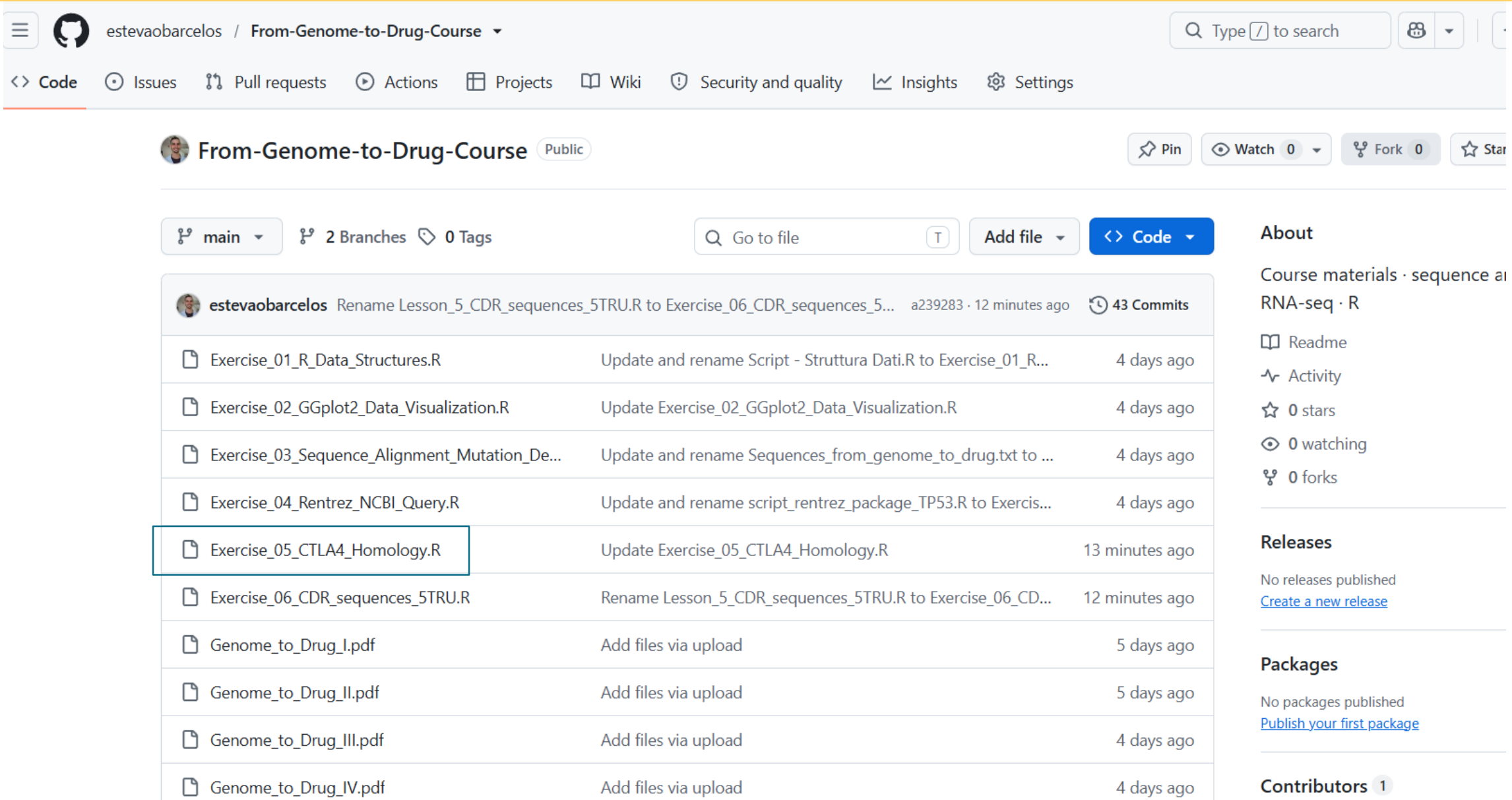


Exercise_05_Homology

CTLA4_HUMAN


UniProt
BLAST Align Peptide search ID mapping SPARQL
UniProtKB
CTLA4
Advanced | List
Search
Help
Status
Reviewed (Swiss-Prot) (126)
Unreviewed (TrEMBL) (2,457)
Popular organisms
Human (44)
Mouse (41)
Bovine (34)
Rat (31)
C. elegans (6)
Taxonomy
Filter by taxonomy
Group by
Taxonomy
Keywords
Gene Ontology
UniProtKB 2,583 results or search "CTLA4" as a Gene Name or Protein Name
Tools Download (3k) Add View: Cards Table Customize columns Share
Entry Entry Name Protein Names Gene Names Organism Length
P16410 CTLA4_HUMAN Cytotoxic T-lymphocyte protein 4[...] CTLA4, CD152 Homo sapiens (Human) 223 AA
P09793 CTLA4_MOUSE Cytotoxic T-lymphocyte protein 4[...] Ctla4, Cd152 Mus musculus (Mouse) 223 AA
Q9XSI1 CTLA4_CANLF Cytotoxic T-lymphocyte protein 4[...] CTLA4 Canis lupus familiaris (Dog) (Canis familiaris) 223 AA
P42072 CTLA4_RABIT Cytotoxic T-lymphocyte protein 4[...] CTLA4 Oryctolagus cuniculus (Rabbit) 223 AA
Q9MYX7 CTLA4_PIG Cytotoxic T-lymphocyte protein 4[...] CTLA4 Sus scrofa (Pig) 223 AA
Q96CW1 AP2M1_HUMAN AP-2 complex subunit mu[...] AP2M1, CLAPM1, KIAA0109 Homo sapiens (Human) 435 AA
O75144 ICOSL_HUMAN ICOS ligand[...] ICOSLG, B7H2, B7RP1, ICOSL, KIAA0653 Homo sapiens (Human) 302 AA
P33681 CD80_HUMAN T-lymphocyte activation antigen CD80[...] CD80, CD28LG, CD28LG1, LAB7 Homo sapiens (Human) 288 AA
Help

https://github.com/estevaobarcelos/from-genome-to-drug-course



RStudio

Source on Save | Run | Sour

```
# ============================================================
#  CTLA-4 Homology Exercise — complete script
#  Species: Human, Mouse, Rabbit, Pig
# ============================================================

BiocManager::install() # Biostrings
BiocManager::install() # msa

install.packages() # ape
install.packages() # pheatmap
install.packages() # bio3d
install.packages() # seqinr

# --- 0. Load libraries ------------------------------------
library(Biostrings)
library(msa)
library(ape)
library(pheatmap)
library(bio3d)
library(seqinr)

# --- 1. Fetch sequences from UniProt ----------------------
fetch_seq <- function(acc) {
  url <- paste0("https://rest.uniprot.org/uniprotkb/", acc, ".fasta")
  readAAStringSet(url)
}

human  <- fetch_seq("P16410")
mouse  <- fetch_seq("P09793")
rabbit <- fetch_seq("P42072")
pig    <- fetch_seq("Q9MYX7")

ctla4 <- c(human, mouse, rabbit, pig)
names(ctla4) <- c("Human", "Mouse", "Rabbit", "Pig")

# Check lengths — should all be ~223
width(ctla4)


# --- 2. Multiple sequence alignment -----------------------
```

UniProtKB accession numbers

RStudio

use Clustal Omega via R to align the sequences

```r
# --- 2. Multiple sequence alignment -------------------------
aln <- msa(ctla4, method = "ClustalOmega")
print(aln, show = "complete")


# --- 3. Pairwise % identity matrix --------------------------
aln_mat <- as.matrix(aln)
n       <- nrow(aln_mat)
nms     <- rownames(aln_mat)
pid_mat <- matrix(0, n, n, dimnames = list(nms, nms))

for (i in 1:n) {
  for (j in 1:n) {
    same  <- sum(aln_mat[i,] == aln_mat[j,] & aln_mat[i,] != "-")
    total <- sum(aln_mat[i,] != "-" | aln_mat[j,] != "-")
    pid_mat[i,j] <- round(same / total * 100, 1)
  }
}

print(pid_mat)
```

RStudio

Examine the length and the amino acid sequence

```
> ctla4 <- c(human, mouse, rabbit, pig)
> names(ctla4) <- c("Human", "Mouse", "Rabbit", "Pig")
> # Check lengths - should all be ~223
> width(ctla4)
[1] 223 223 223 223
>
```

```
> print(aln, show = "complete")

MsaAAMultipleAlignment with 4 rows and 223 columns
    aln (1..91)                                                                 names
[1] MACLGLRRYKAQLQLPSRTWPFVALLTLLFIPVFSEAIQVTQPSVVLASSHGVASFPCEYSPSHNTDEVRVTVLRQTNDQMTEVCATTFTE Mouse
[2] MARLGFQRQGTQLDLASRTWSCAALFSLLFLPVFSKALHVSQPAVVLASSRGVASFVCEYASSHKATEVRVTVLRQANSQMTEVCAMTYTV Rabbit
[3] MACLGFQRHKAQLNLATRTWPCTLLFFLLFIPVFCKAMHVAQPAVVLASSRGIASFVCEYASPGKATEVRVTVLRQADSQVTEVCAATYMM Human
[4] MACSGFQSHGAWLELTSRTWPCTALFSLLFIPVFSKGMHVAQPAVVLANSRGVASFVCEYGSAGKAAEVRVTVLRRAGSQMTEVCAATYTV Pig
Con MACLGFQRH?AQL?LASRTWPCTALFSLLFIPVFSKAMHVAQPAVVLASSRGVASFVCEYASS?KATEVRVTVLRQANSQMTEVCAATYTV Consensus

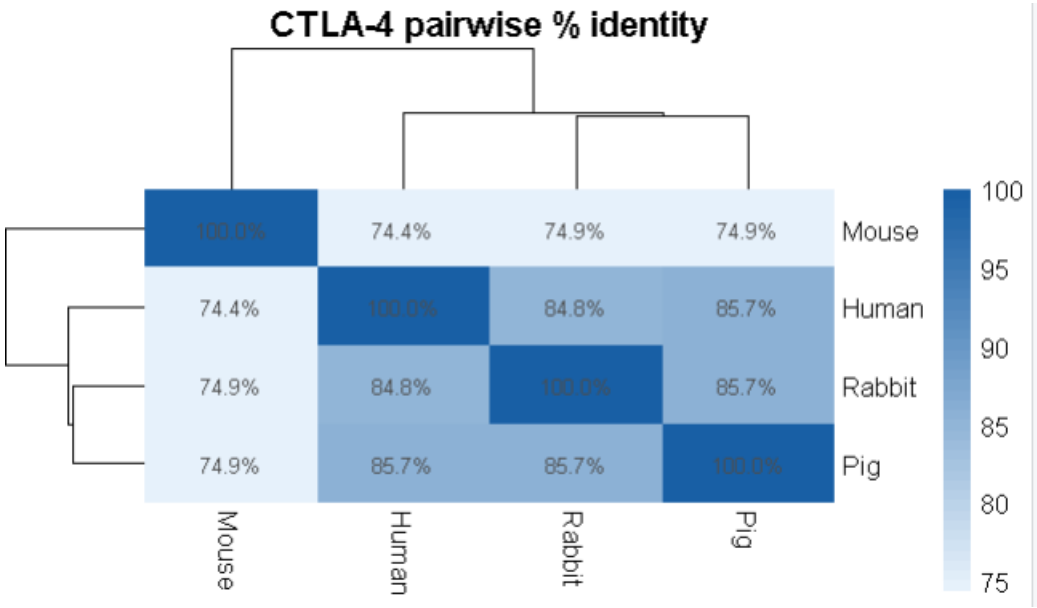
    aln (92..182)                                                               names
[1] KNTVGFLDYPFCSGTFNESRVNLTIQGLRAVDTGLYLCKVELMYPPPYFVGMGNGTQIYVIDPEPCPDSDFLLWILVAVSLGLFFYSFLVS Mouse
[2] ENELTFIDDSTCTGISHGNKVNLTIQGLSAMDTGLYICKVELMYPPPYYVGMGNGTQIYVIEPEPCPDSDFLLWILAAISSGLFFYSFLIT Rabbit
[3] GNELTFLDDSICTGTSSGNQVNLTIQGLRAMDTGLYICKVELMYPPPYYLGIGNGTQIYVIDPEPCPDSDFLLWILAAVSSGLFFYSFLLT Human
[4] EDELTFLDDSTCTGTSTENKVNLTIQGLRAVDTGLYICKVELLYPPPYYVGMGNGTQIYVIDPEPCPDSDFLLWILAAVSSGLFFYSFLIT Pig
Con ENELTFLDDSTCTGTS??NKVNLTIQGLRA?DTGLYICKVELMYPPPYYVGMGNGTQIYVIDPEPCPDSDFLLWILAAVSSGLFFYSFLIT Consensus

    aln (183..223)                          names
[1] AVSLSKMLKKRSPLTTGVYVKMPPTEPECEKQFQPYFIPIN Mouse
[2] AVSLSKMLKKRSPLTTGVYVKMPPTEPECEKQFQPYFIPIN Rabbit
[3] AVSLSKMLKKRSPLTTGVYVKMPPTEPECEKQFQPYFIPIN Human
[4] AVSLSKMLKKRSPLTTGVYVKMPPTEPECEKQFQPYFIPIN Pig
Con AVSLSKMLKKRSPLTTGVYVKMPPTEPECEKQFQPYFIPIN Consensus
>
```

RStudio

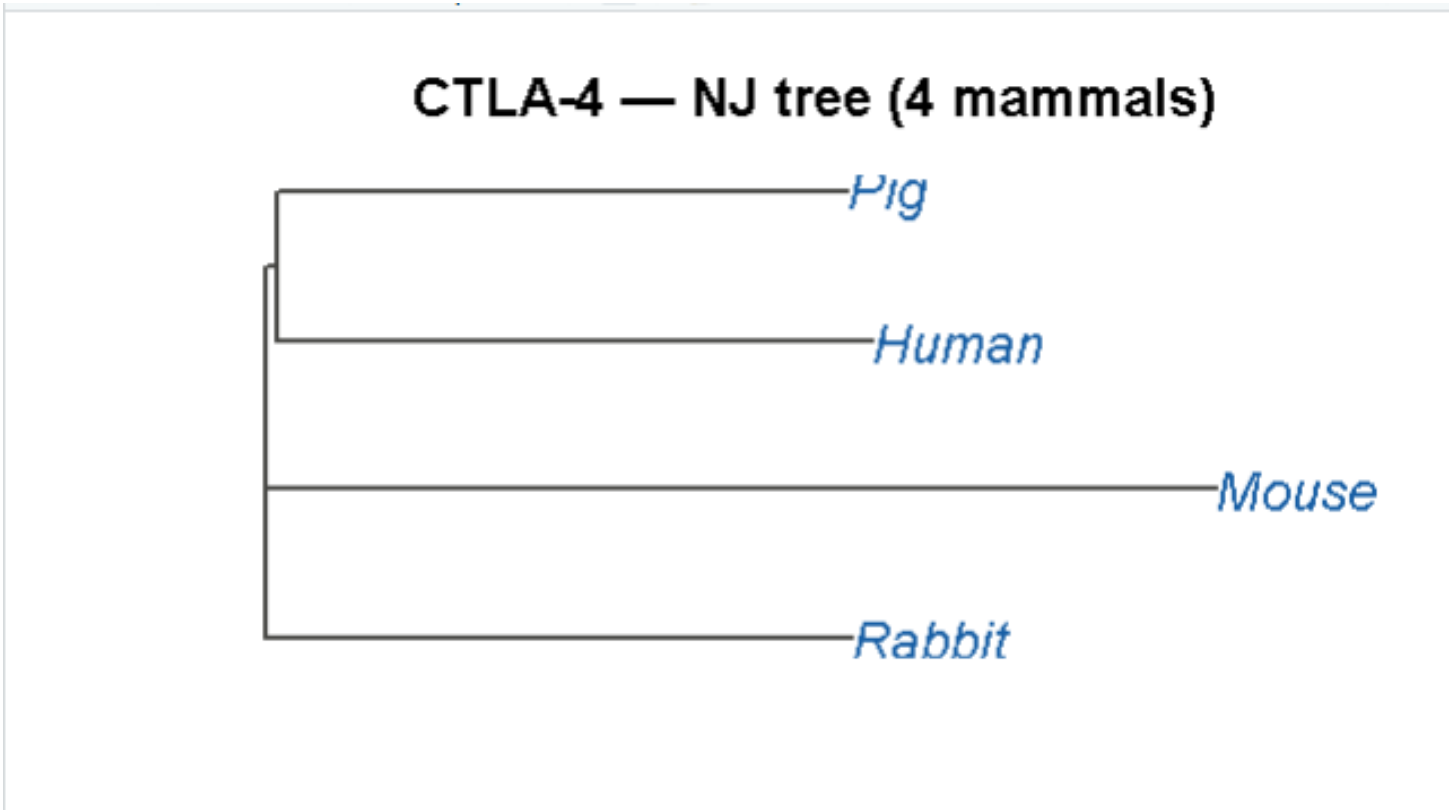
Examine pairwise identities

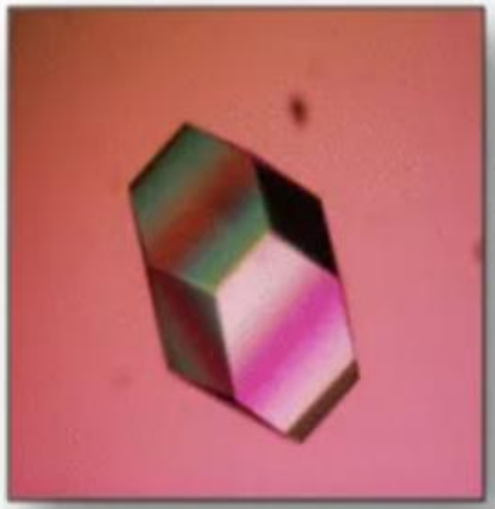
```
> print(pid_mat)
       Mouse Rabbit Human   Pig
Mouse  100.0   74.9  74.4  74.9
Rabbit  74.9  100.0  84.8  85.7
Human   74.4   84.8 100.0  85.7
Pig     74.9   85.7  85.7 100.0
>
```



RStudio

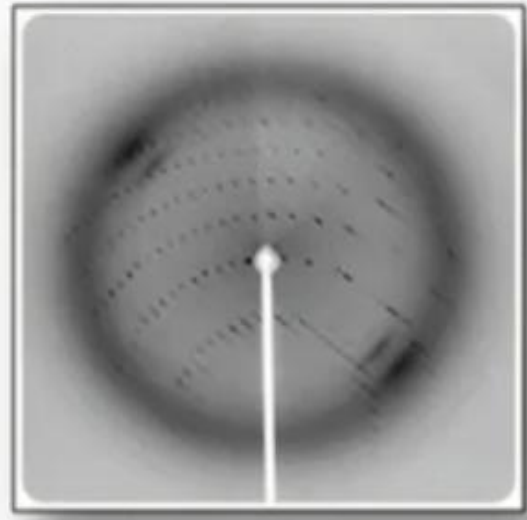
Neighbor-Joining Tree





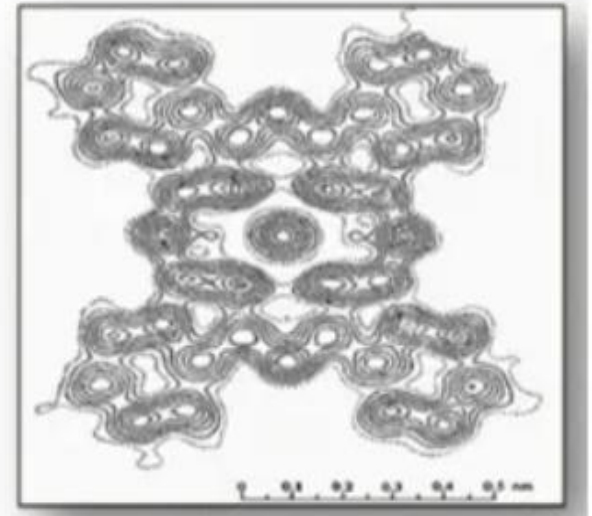
Protein crystal

X-ray diffraction



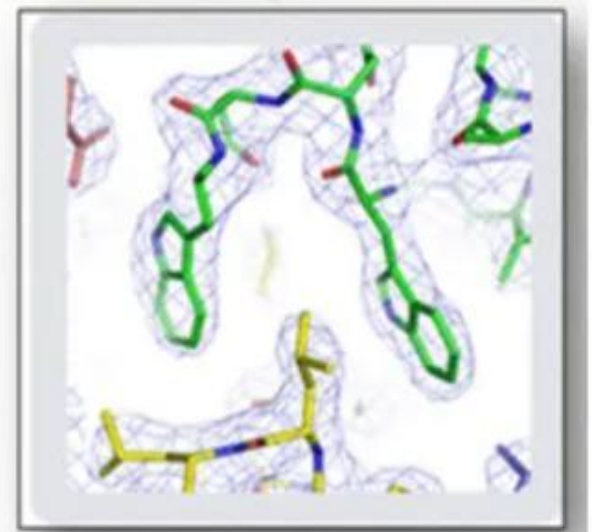
Diffraction pattern

Phasing

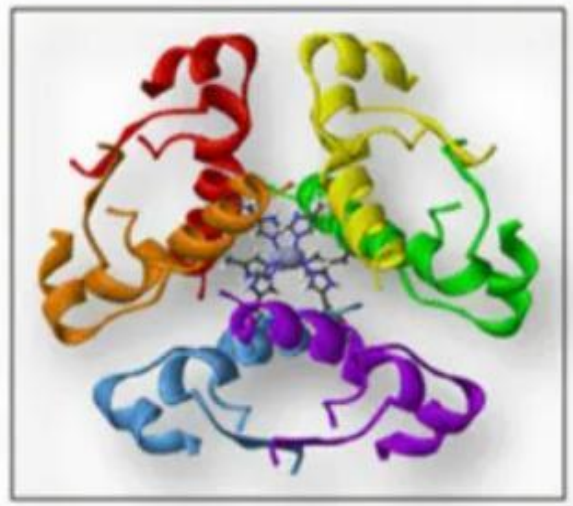


Electron density map

Molecular modeling

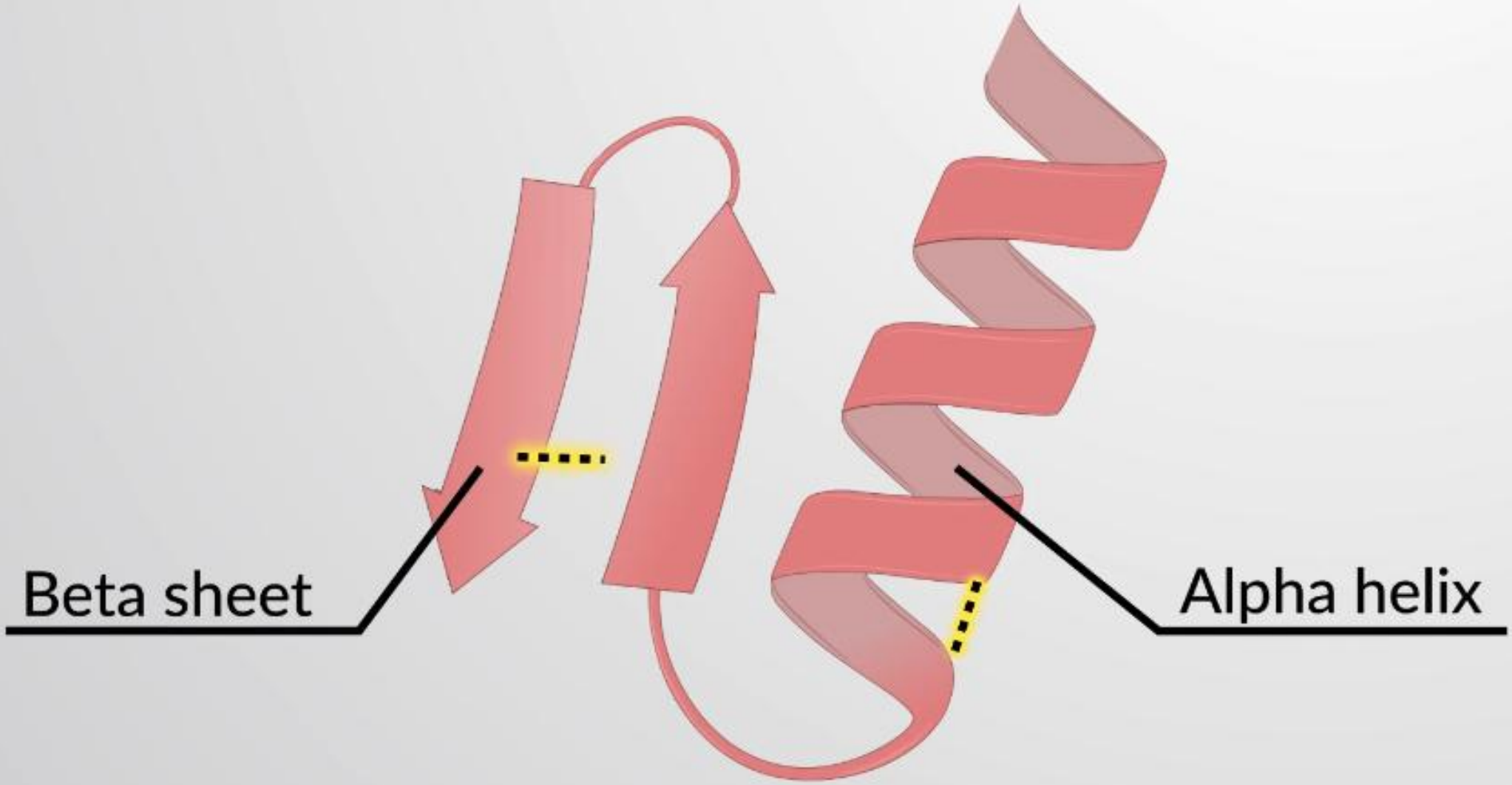


Model evaluation /
PDB submission



Final model structure

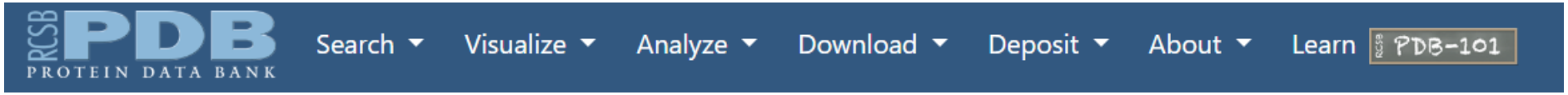
Secondary structure of a protein



What Is the PDB?

The **RCSB Protein Data Bank (PDB)** at rcsb.org is the:

- world's primary open-access repository for 3D structural data of biological macromolecules — proteins, nucleic acids, and their complexes.
- Each deposited structure receives a unique 4-character alphanumeric accession code (the PDB ID).
- The database currently holds over 220,000 structures determined by X-ray crystallography, NMR spectroscopy, or cryo-electron microscopy.

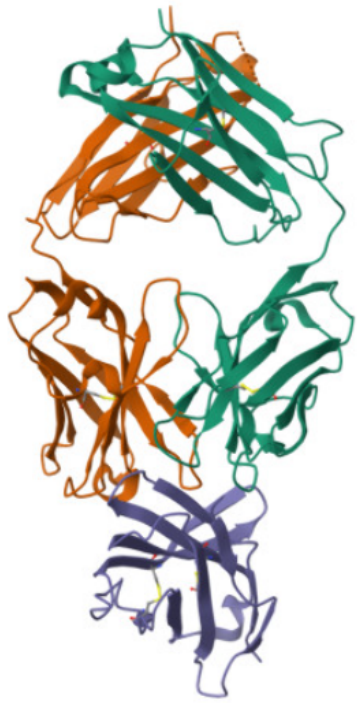


5TRU - Ipilimumab

RCSB PDB PROTEIN DATA BANK | Search | Visualize | Analyze | Download | Deposit | About | Learn | PDB-101

Structure Summary | Structure | Annotations | Experiment | Sequence | Genome | Versions

Biological Assembly 1



Explore in 3D: Structure | Sequence Annotations | Electron Density | Validation Report

Display Files | Download Files | Data API

5TRU | pdb_00005tru

Structure of the first-in-class checkpoint inhibitor Ipilimumab bound to human CTLA-4

PDB DOI: https://doi.org/10.2210/pdb5TRU/pdb

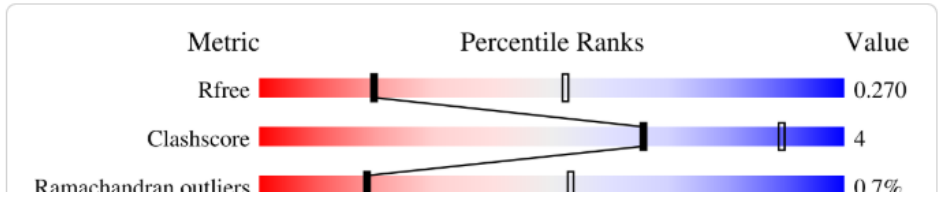
Classification: IMMUNE SYSTEM
Organism(s): Homo sapiens
Expression System: Cricetulus griseus, Escherichia coli BL21(DE3)
Mutation(s): No

Deposited: 2016-10-27 **Released:** 2017-05-10
Deposition Author(s): Ramagopal, U.A., Liu, W., Garrett-Thomson, S.C., Yan, Q., Srinivasan, M., Wong, S.C., Bell, A., Mankikar, S., Rangan, V.S., Deshpande, S., Bonanno, J.B., Korman, A.J., Almo, S.C.

Experimental Data Snapshot

Method: X-RAY DIFFRACTION
Resolution: 3.00 Å
R-Value Free:
0.268 (Depositor), 0.270 (DCC)
R-Value Work:

wwPDB Validation | 3D Report | Full Report

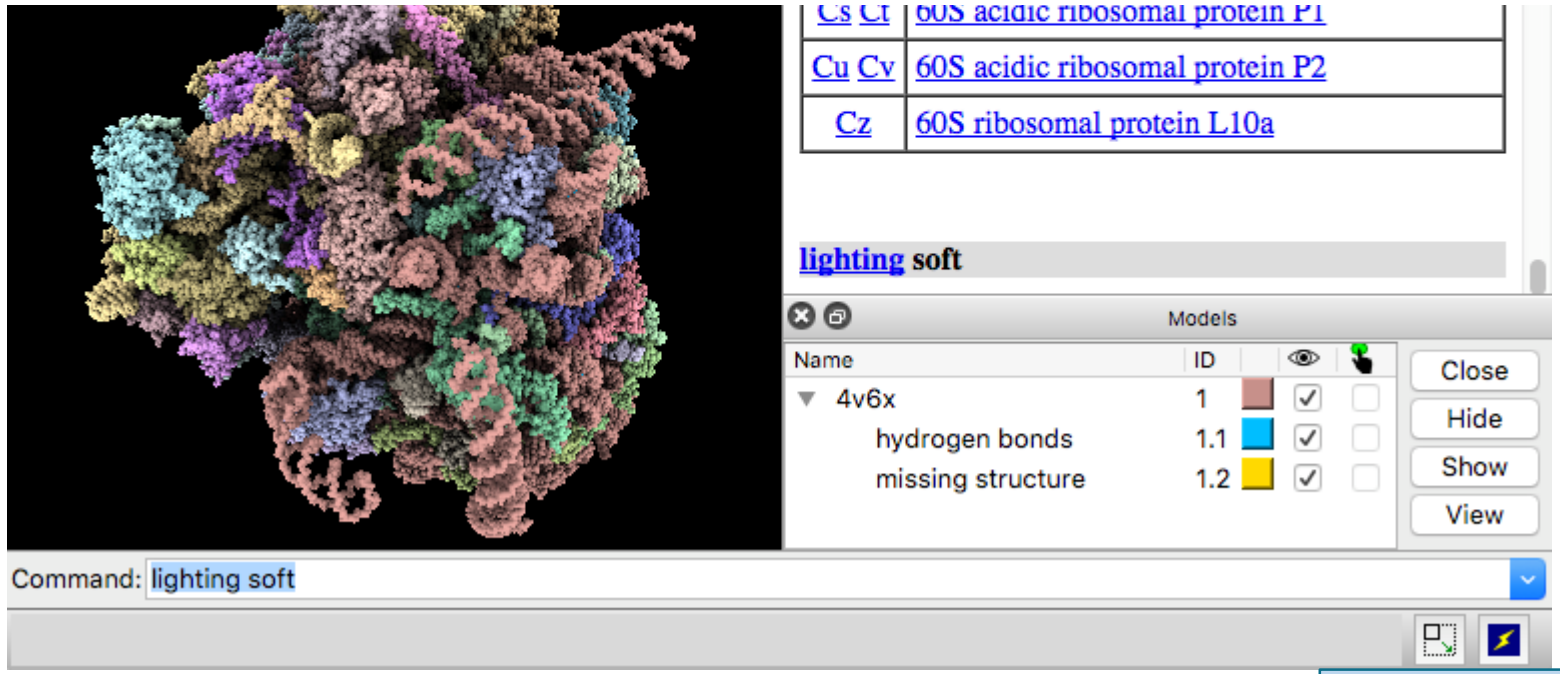


ChimeraX

UCSF RBVI
https://www.cgl.ucsf.edu › downl... · Traduzir esta página

Download UCSF ChimeraX

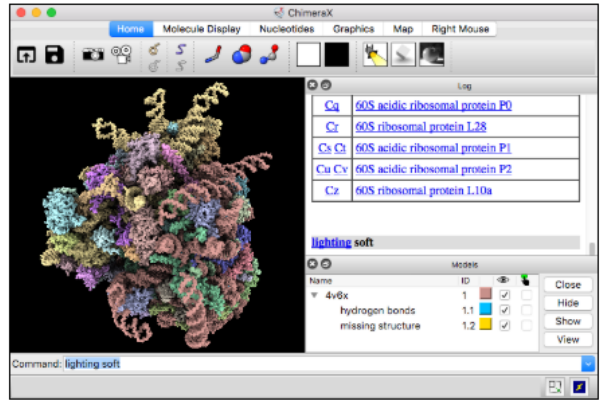
Install on your personal computer with: **sudo flatpak install ~/Downloads/ChimeraX-1.12.flatpak** . Run with: flatpak run edu.ucsf.rbvi.ChimeraX . More Info ...



Exercise Visualization

https://www.cgl.ucsf.edu/chimerax/download.html

Download UCSF ChimeraX



ChimeraX is the state-of-the-art visualization program from the Resource for Biocomputing, Visualization, and Informatics at UC San Francisco. It is free for academic, government, nonprofit, and personal use; commercial users, please see commercial licensing. Please cite ChimeraX in publications.

See also:

- ChimeraX Documentation
- System Requirements
- Change Log
- Download & Citation Counts
- Older Releases
- Common Problems

Current releases:

- Production Builds
- Daily Builds

ChimeraX 1.12

Production releases are stable versions for ChimeraX Toolshed bundles to work with. You may need to use an older release if a bundle you wish to use has not been updated yet. Showing releases for Windows 10.

Operating System	Distribution	Size	Date	Notes
Windows (x86-64)	ChimeraX-1.12.exe	391.88 MiB	11 giu 2026	Download is a Windows (Inno Setup based) installer. Tested on Windows 10 and Windows 11. ▶ More Info...

▶ Other releases

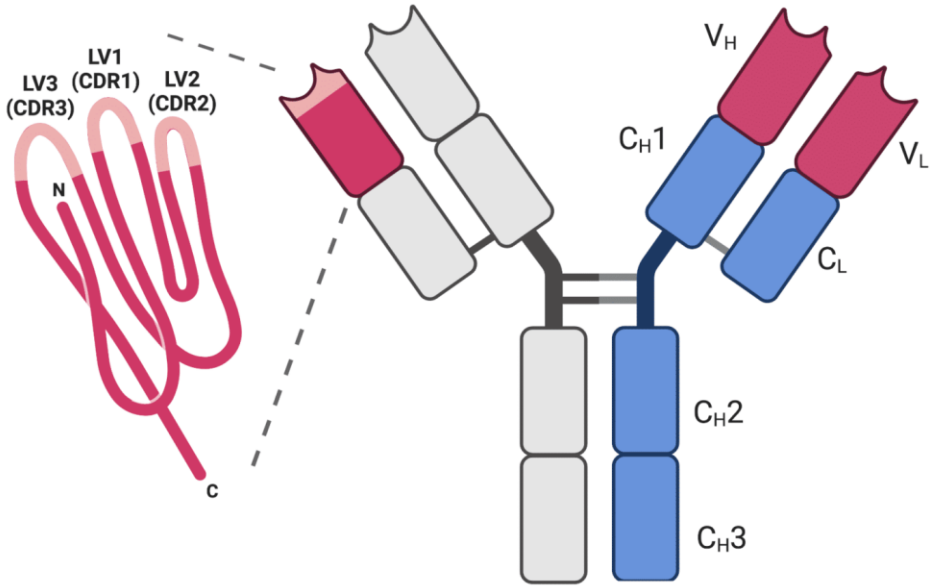
Daily Build

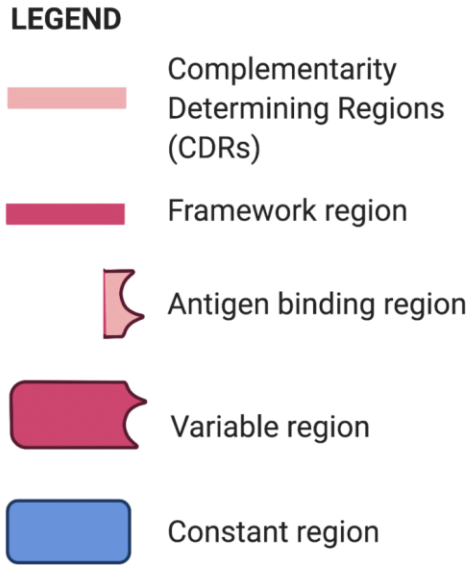
Daily builds are generated automatically each night from the development source code (see the change log). While a given build may have unforeseen problems, these are often fixed by the next day. Showing releases for Windows 10.

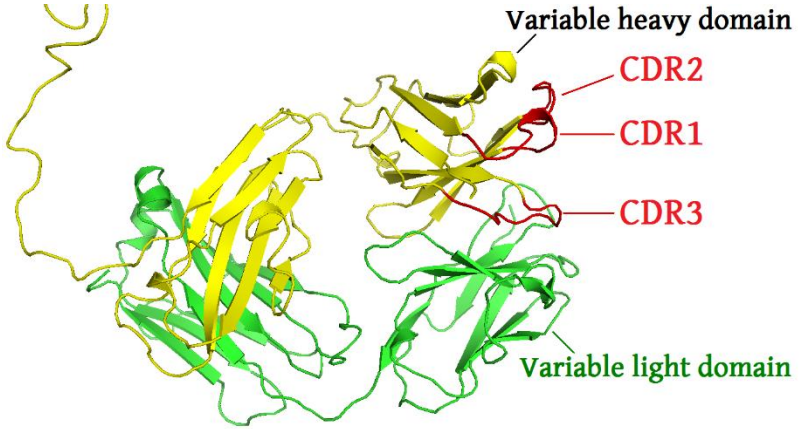
Operating System	Distribution	Size	Date	Notes
				Download is a Windows (Inno Setup based) installer. Tested on Windows 10 and

Complementarity-Determining Region

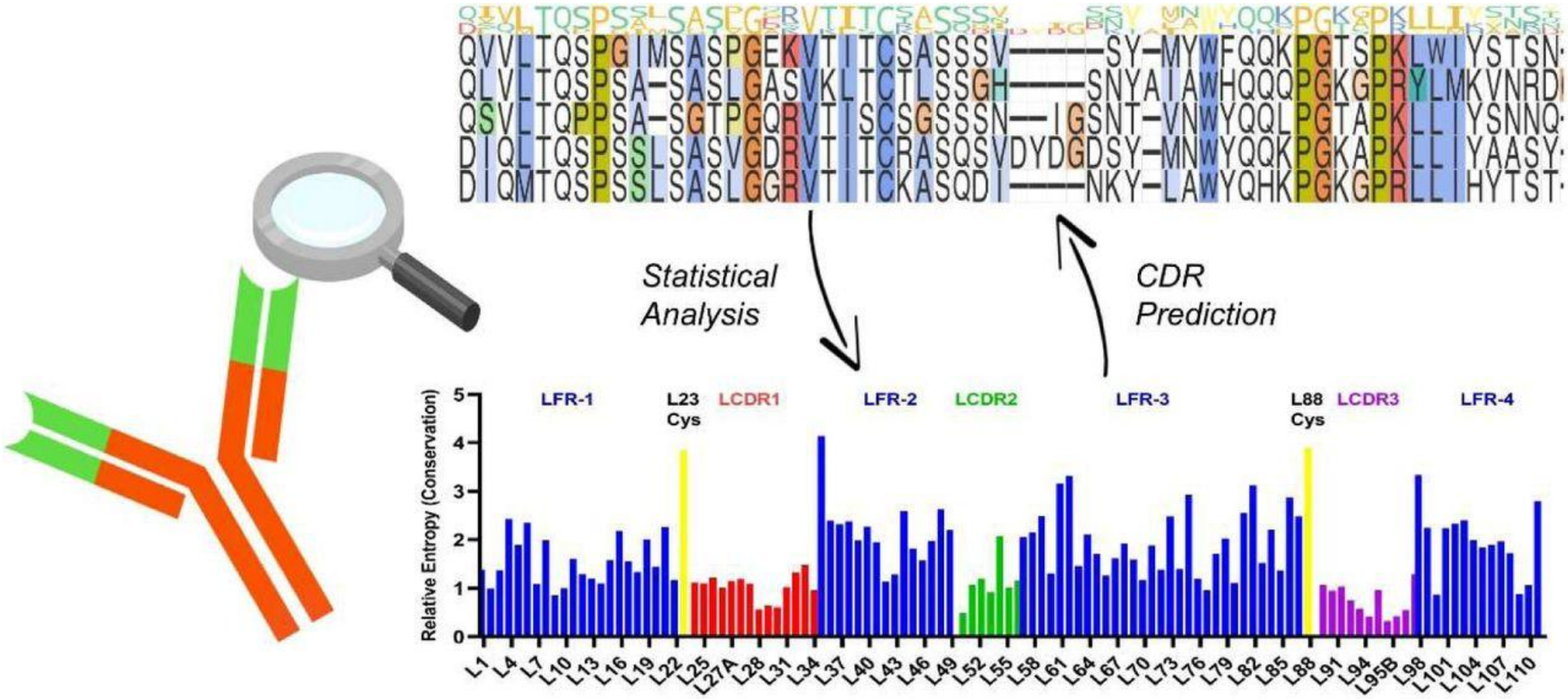
The **complementarity-determining regions (CDRs)** of antibodies are **three hypervariable loops** within the variable domains of the **heavy and light chains** that form the **antigen-binding site** and play a crucial role in **determining antigen specificity and binding affinity**.







The Kabat Numbering Scheme

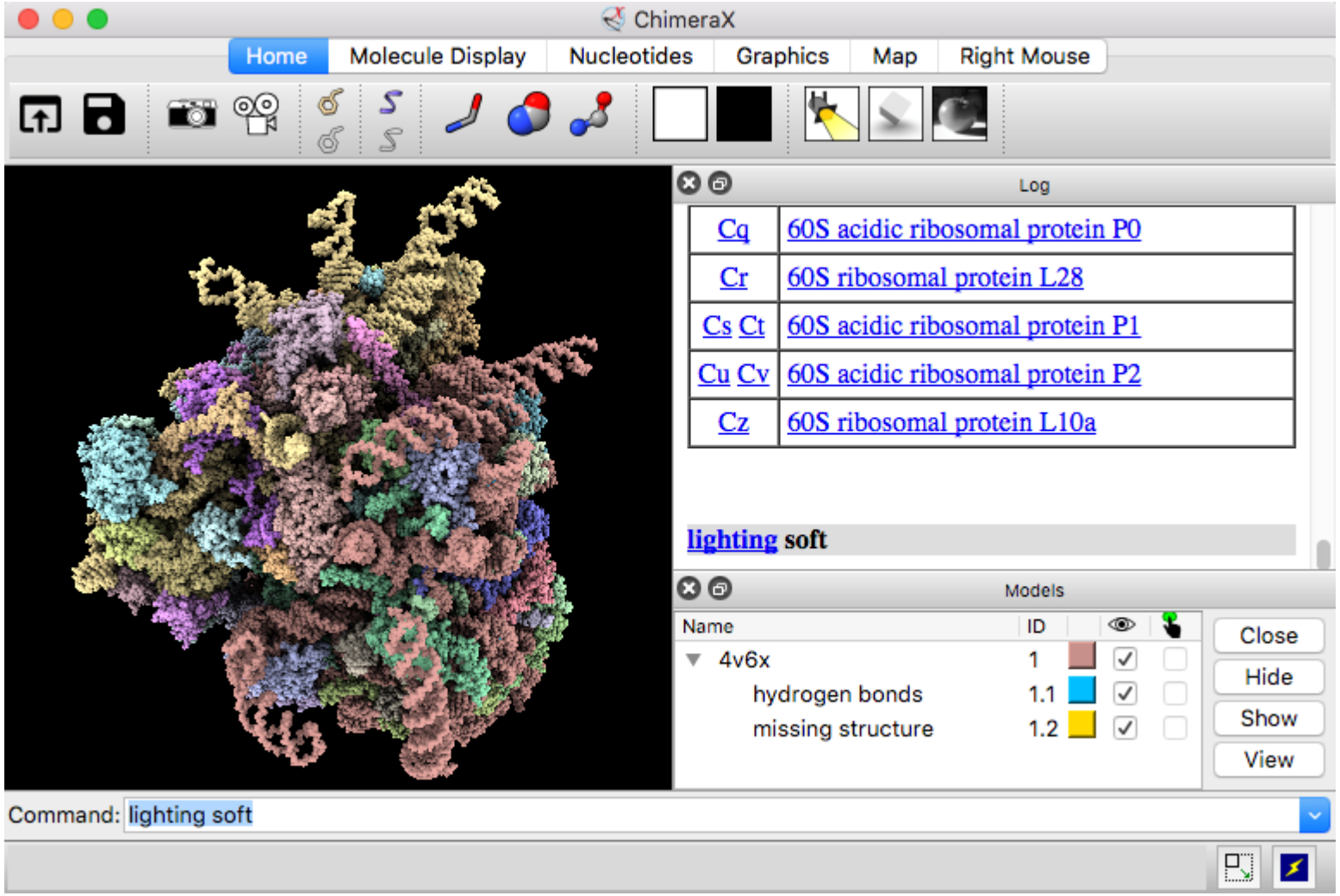


http://www.bioinf.org.uk/abs/info.html

Kabat versus Chothia

CDR	Kabat	Chothia	Martin *	PyIgClassify **	IMGT **
L1	24–34	24–34	24–34	24–34	27–32 (M − 5)
L2	50–56	50–56	50–56	49–56 (M + 1)	50–52 (M − 4)
L3	89–97	89–97	89–97	89–97	89–97
H1	31–35	26–32	26–35 (K + C)	23–35 (M + 3)	26–33 (M − 2)
H2	50–65	52–56	50–58 (K − 7)	50–58	51–57 (M − 2)
H3	95–102	95–102	95–102	93–102 (M + 2)	93–102 (M + 2)

R + ChimeraX



Exercise Visualization

RStudio

Extract CDR sequences from 5TRU

```r
# --- 0. Load libraries ------------------------------------
library(bio3d)


# --- 1. Download structure --------------------------------
pdb <- read.pdb("5TRU")

# Check chains
unique(pdb$atom$chain)
# H = heavy chain, L = light chain, A = CTLA-4


# --- 2. Extract CDR sequences (Kabat numbering) -----------

# Function to get sequence of a selected region
get_seq <- function(pdb, chain, resno_range) {
  sel <- atom.select(pdb,
                     chain = chain,
                     resno = resno_range,
                     elety = "CA")
  res <- pdb$atom$resid[sel$atom]
  aa  <- aa321(res)        # convert 3-letter to 1-letter code
  paste(aa, collapse = "")
}
```

RStudio

Extract CDR sequences from 5TRU

```
# Heavy chain CDRs
CDR_H1 <- get_seq(pdb, "H", 31:35)
CDR_H2 <- get_seq(pdb, "H", 50:65)
CDR_H3 <- get_seq(pdb, "H", 95:102)

# Light chain CDRs
CDR_L1 <- get_seq(pdb, "L", 24:34)
CDR_L2 <- get_seq(pdb, "L", 50:56)
CDR_L3 <- get_seq(pdb, "L", 89:97)

# --- Print all CDRs ----------------------------------------
cat("=== Ipilimumab CDR sequences (from PDB 5TRU) ===\n\n")
cat("Heavy chain:\n")
cat("  CDR-H1 (26-35) :", CDR_H1, "\n")
cat("  CDR-H2 (50-65) :", CDR_H2, "\n")
cat("  CDR-H3 (95-102):", CDR_H3, "\n\n")
cat("Light chain:\n")
cat("  CDR-L1 (24-34) :", CDR_L1, "\n")
cat("  CDR-L2 (50-56) :", CDR_L2, "\n")
cat("  CDR-L3 (89-97) :", CDR_L3, "\n")
```

RStudio

Get full heavy and light chain sequences

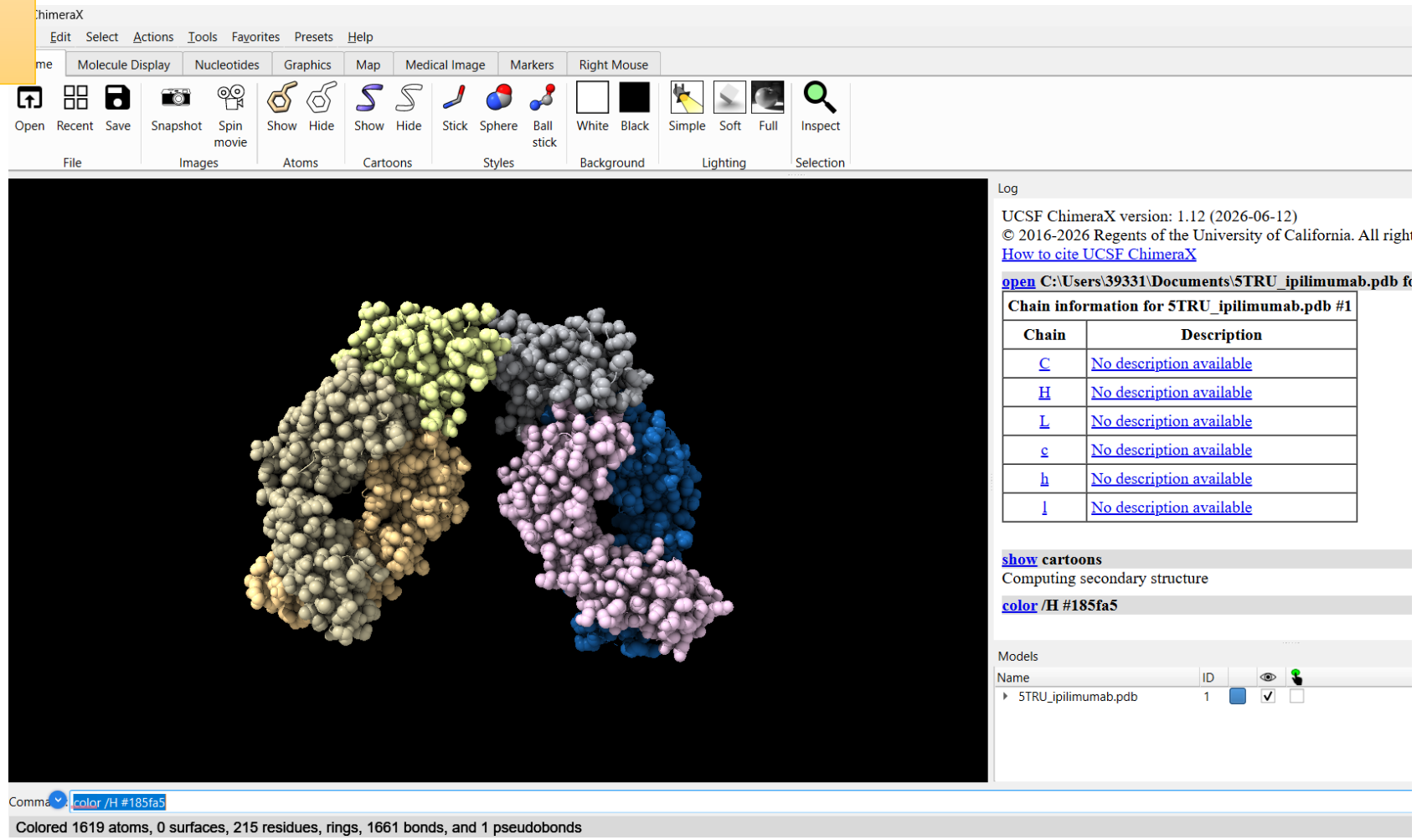
```
54  get_full_seq <- function(pdb, chain) {
55    sel <- atom.select(pdb, chain = chain, elety = "CA")
56    res <- pdb$atom$resid[sel$atom]
57    # keep unique residues in order
58    resno <- pdb$atom$resno[sel$atom]
59    idx   <- !duplicated(resno)
60    aa    <- aa321(res[idx])
61    paste(aa, collapse = "")
62  }
63
64  seq_H <- get_full_seq(pdb, "H")
65  seq_L <- get_full_seq(pdb, "L")
66  seq_A <- get_full_seq(pdb, "C")
67
68  cat("\n=== Full sequences ===\n")
69  cat("Heavy chain :", nchar(seq_H), "aa\n", seq_H, "\n\n")
70  cat("Light chain :", nchar(seq_L), "aa\n", seq_L, "\n\n")
71  cat("CTLA-4      :", nchar(seq_A), "aa\n", seq_A, "\n\n")
72
73
74  # --- 4. Save PDB for ChimeraX ------------------------------
75  # Save the full structure
76  write.pdb(pdb, file = "5TRU_ipilimumab.pdb")
77  cat("PDB saved as: 5TRU_ipilimumab.pdb\n")
78  cat("Working directory:", getwd(), "\n")
79
```

Save the file

ChimeraX Program

Open the file into ChimeraX

File > open > found the file .pdb

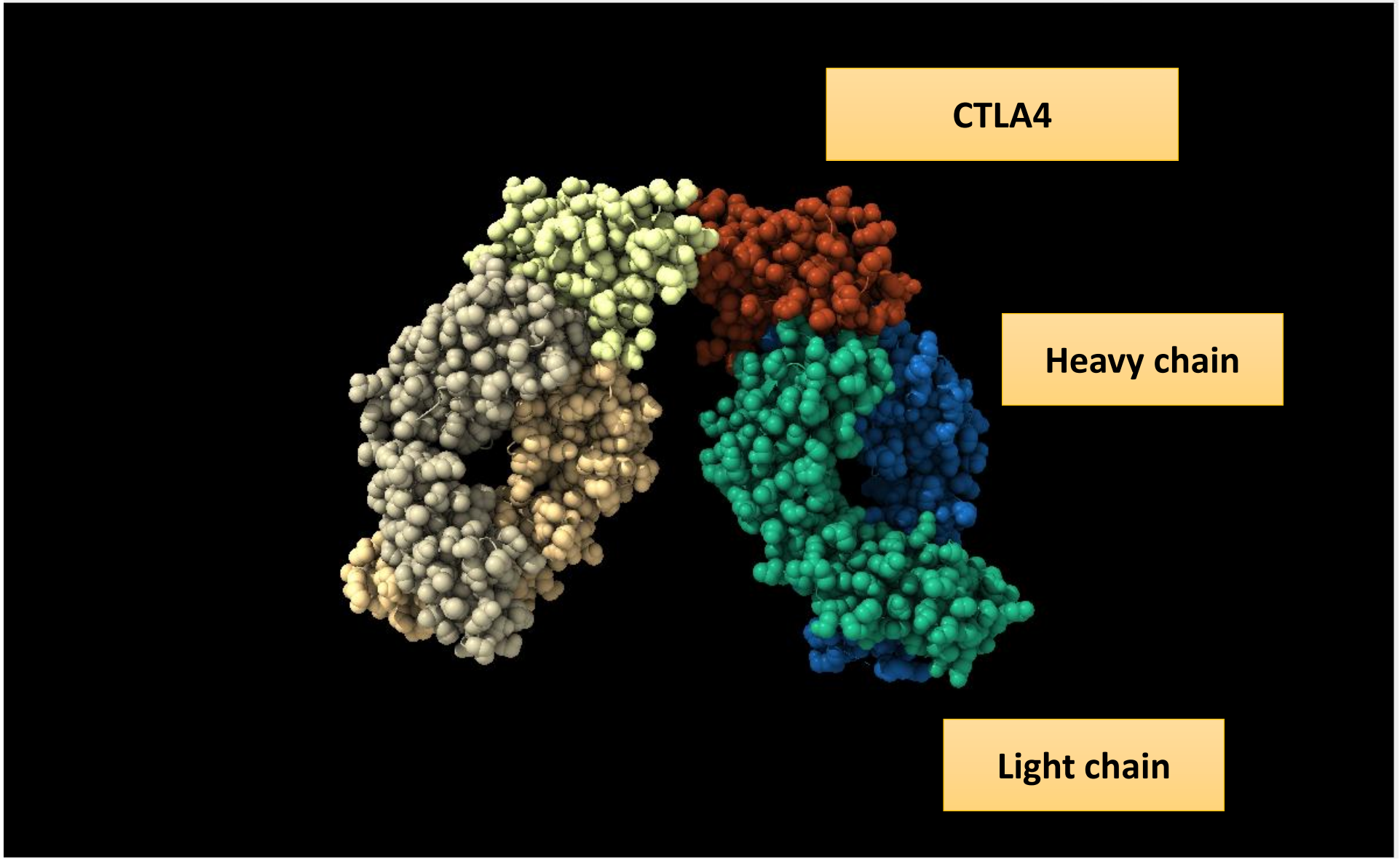


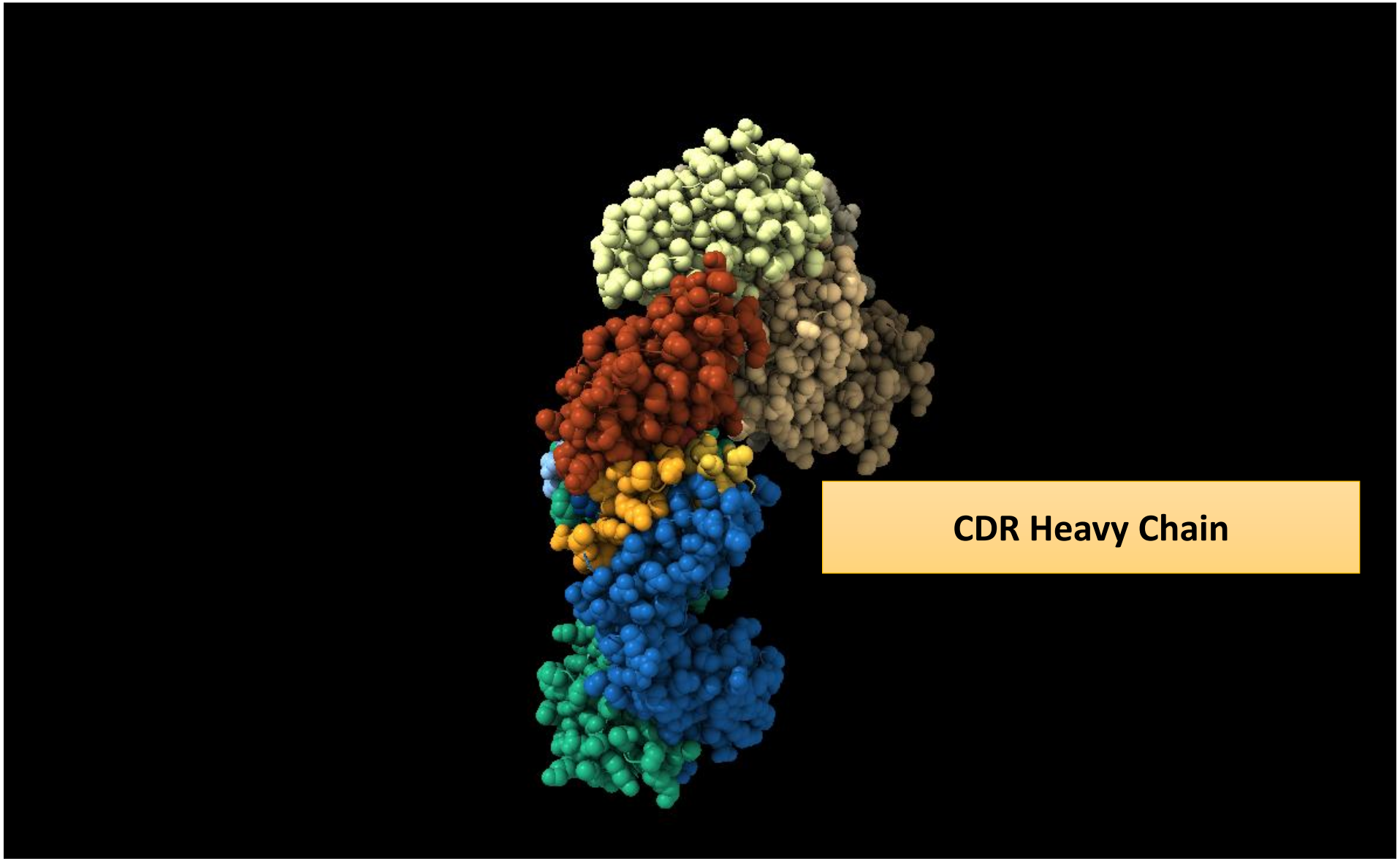
All commands need to be pasted here.

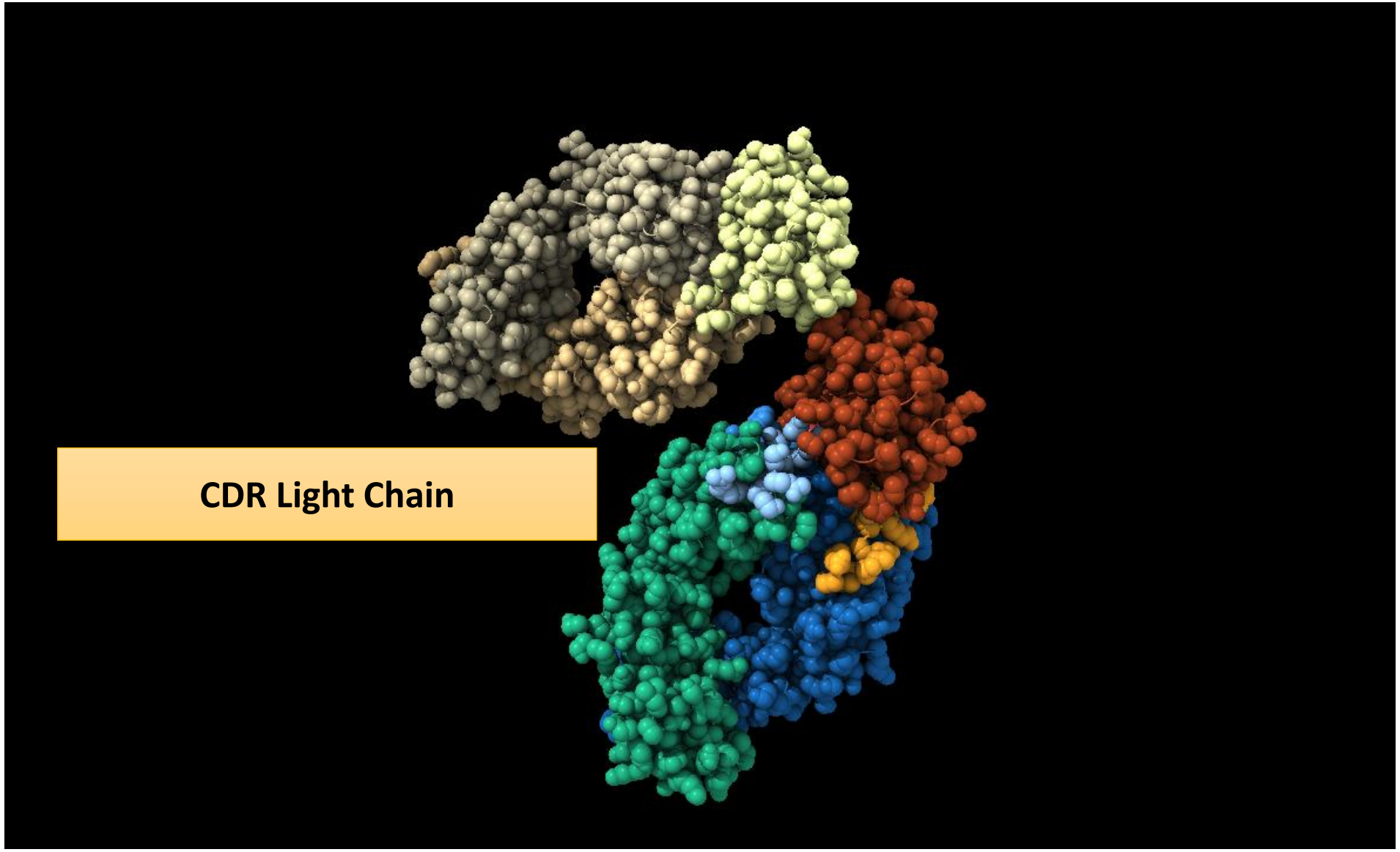
ChimeraX Program

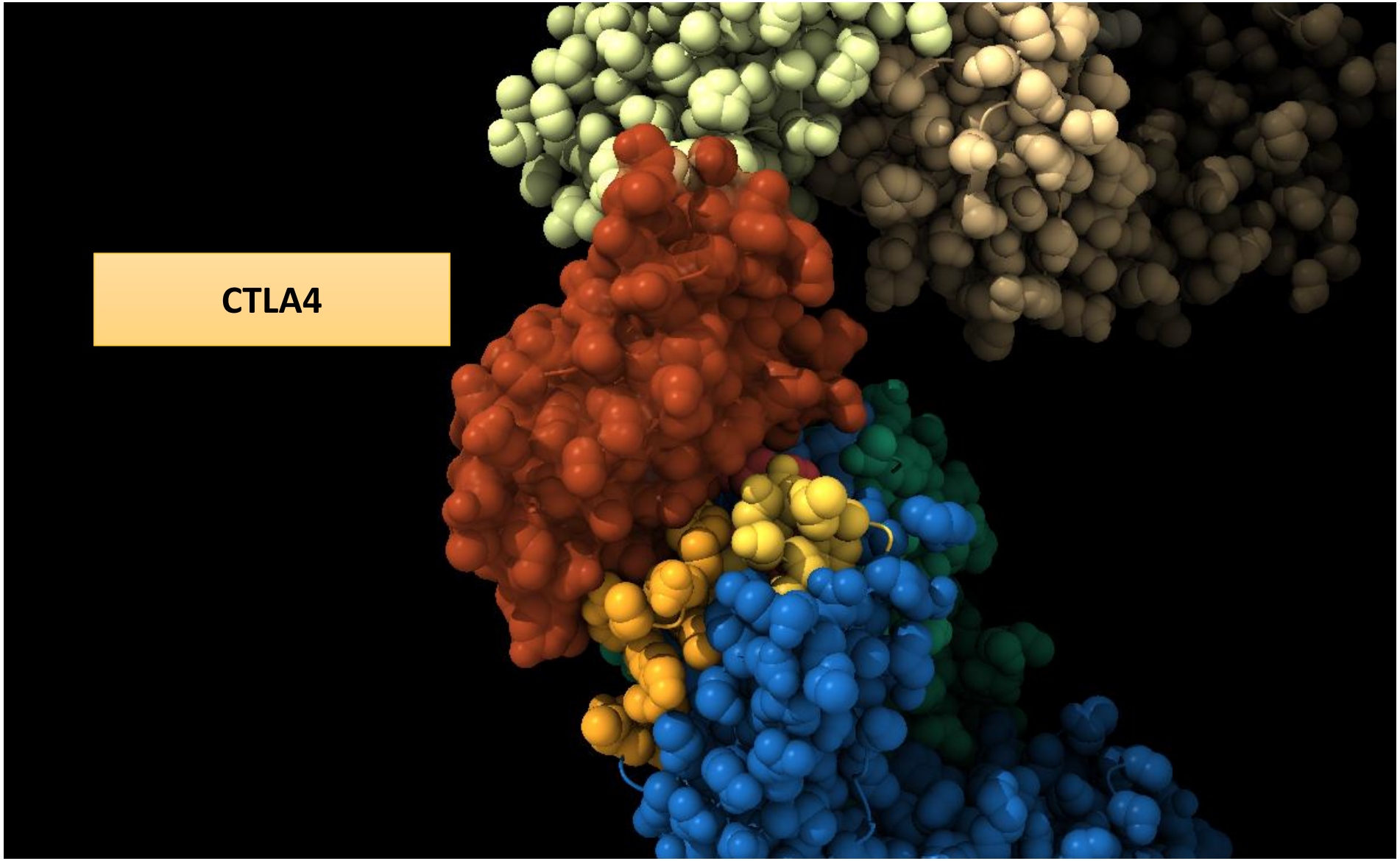
Distinguish the chains and Visualize the CDR regions

```
 95  # 2. Color the 3 chains clearly
 96  color /H #185fa5
 97  color /L #1d9e75
 98  color /A #993c1d
 99
100  # 3. Highlight the 6 CDR loops
101  color /H:26-35  #f5c842
102  color /H:50-65  #ef9f27
103  color /H:95-102 #e24b4a
104  color /L:24-34  #85b7eb
105  color /L:50-56  #378add
106  color /L:89-97  #0c447c
107
```


CTLA4
Heavy chain
Light chain


CDR Heavy Chain


CDR Light Chain


CTLA4